RESEARCH STATEMENT

# Understanding and Improving AI Safety through Internal Representations

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**I develop mechanistic methods to diagnose, shape, and control the safety behavior of foundation models and AI agents.** My thesis is that reliable safety requires connecting observable behavior to the internal mechanisms that sustain it under perturbation and adaptation. A correct answer can conceal fragile reasoning; a refusal can conceal an accessible path to harmful behavior. These gaps make representation-level evaluation and intervention central to my research agenda.

My laboratory will establish **mechanistic foundations for monitoring and steering**: identify safety-relevant computation, determine when it predicts emerging risk, and use it to intervene while preserving useful capabilities. I bring together representation-level safety evaluation, alignment, and adversarial robustness to pursue this agenda. Robustness is my methodological foundation; AI safety is my central scientific focus; healthcare and climate provide selective settings for studying consequential distribution shift.

## 1. Current Research: Diagnose, Shape, and Intervene

I originated these projects and drove their experimental development as first or co-first author. They connect three themes: diagnosing what evaluation misses, enriching alignment signals, and reducing the cost of safety intervention.

**Diagnose: Reveal vulnerabilities that behavior alone can miss.** In *MATCHA* [1], I introduced answer-conditioned probing to test whether answer accuracy captures reasoning robustness. Our experiments show that correct answers can coexist with inconsistent rationales, with multi-step and commonsense tasks more susceptible than logical tasks. In *When Behavioral Safety Evaluation Fails* [2], my collaborators and I demonstrate a complementary *audit gap*: models that resemble their aligned bases under static audits can be substantially more vulnerable internally. Across three model families, constructed models exhibit **2.5–3.1 times the base models' Latent Vulnerability Score** at the targeted middle layer, measuring safety degradation per unit of bounded latent perturbation. The central finding is a mismatch between what these metrics measure and the robustness they are intended to establish.

**Shape: Enrich the learning signal for alignment.** I proposed *GANPO* [3] to give preference learning richer feedback through internal representations. Its latent adversarial regularizer penalizes divergence between policy and reference representations, complementing token-level constraints. Across evaluated architectures and tasks, latent regularization improves preference optimization; under distribution shift and noise, it provides more robust structural feedback with comparable downstream performance and minor computational overhead. This shows that internal representations supply useful alignment signals beyond token-space similarity.

**Intervene: Reduce the alignment tax through selective control.** I co-led *CLEAR* [4], accepted to the Trustworthy AI for Good Workshop at NeurIPS 2026, which uses a hidden-state gate to control when a safety adapter intervenes. On Llama-3-8B-Instruct, CLEAR reduces HarmBench attack success from 32.3% to 0.5% while achieving up to 7.1 percentage points higher GSM8K accuracy than globally applied safety tuning or LoRA. By making selective intervention part of alignment itself, this work motivates my future research on risk-sensitive monitoring and steering across agent trajectories.

## 2. Methodological Foundations: Robustness and Distribution Shift

**Robustness under explicit threat models.** My work on *RAMP* [5] addresses competing objectives in multi-norm adversarial training; *CURE* [6] extends this investigation to certified training. In federated domain adaptation [7], my collaborators and I develop a theoretical framework for aggregation under domain shift and limited target data, yielding gradient projection and an auto-weighting rule. Together, these projects supply the technical foundation for my safety agenda: explicit uncertainty models, principled optimization, and careful distinctions between empirical protection and certified guarantees.

**Selective applications in healthcare and AI for Science.** My federated learning research is motivated in part by heterogeneous healthcare data. In *PyroAdapt* [8], I develop a pretrain–retrieve–rank framework for wildfire prediction under spatial heterogeneity and temporal shift; it raises daily average precision from 21.62% to 24.57% and improves recall of the most extreme fires under a fixed detection budget. These settings inform my study of consequential distribution shift and offer selective collaborations for testing reliable learning methods with domain-specific outcomes.

## 3. Future Directions: Mechanistic Foundations for Monitoring and Steering

My independent research program will connect two questions: **when does internal state provide actionable warning of unsafe behavior, and which causal mechanisms make an intervention reliable?** This moves my work from diagnosing and shaping individual model responses toward understanding safety over trajectories of reasoning, action, and adaptation. The two thrusts share models and evaluation tools, while each addresses a distinct scientific question.

**Thrust I: Detect and redirect unsafe agent trajectories.** My goal is to detect emerging risk before an agent takes a consequential unsafe action. I will combine internal states with action histories to determine when they provide earlier warning than output-only evaluation. I will test this hypothesis in controlled tool-use environments with explicit task objectives and safety constraints. Matched trajectories will separate successful task progress from unsafe behavior, while held-out tasks and adversarial changes to tool outputs will test generalization. Comparisons will include action-based monitors and static probes at matched observation and compute budgets.

*Initial focus: Single-agent detection.* I will establish a reproducible trajectory benchmark and identify when safety-relevant signals emerge across reasoning and action steps. A public evaluation protocol will support reproducible comparisons of temporal monitors and their limits on held-out conditions. Success will be measured by detection lead time and missed unsafe actions at fixed false-alarm rates, alongside task completion. Calibration across tasks will establish the conditions under which these warnings remain reliable.

*Building on detection: Closed-loop steering.* I will use monitoring signals to select among targeted activation interventions, constrained replanning, and additional verification before consequential actions. Controlled interventions will distinguish causal improvements from changes that merely suppress monitor signals. The scientific objective is to identify when selective intervention improves the safety–utility tradeoff over both unmodified agents and interventions applied at every step. Independent outcome checks will measure safety, task success, unnecessary intervention, and inference cost; rollback experiments will test whether benefits persist after an intervention ends.

*Longer-term opportunity: Multi-agent interaction.* I aim to understand how safety failures propagate through communication and delegation. Controlled interaction settings will test when individual-agent monitors miss risks that emerge across agent boundaries, and whether shared interaction state improves detection. These studies will establish the basis for coordinated steering, with system-level safety and task completion as the evaluation criteria.

**Thrust II: Explain safety mechanisms in open-weight models.** My goal is to explain which computations sustain safe behavior and how alignment and adaptation change them. My working hypothesis is that features associated with refusal and mechanisms that prevent unsafe actions overlap only partially. Access to weights, activations, and model checkpoints makes this distinction experimentally testable.

*Initial focus: Causal characterization.* I will compare models before and after alignment on matched benign and harmful requests, then use activation patching, ablation, and targeted steering to test candidate mechanisms. These causal case studies will report both intervention effects and collateral changes to benign capabilities. Held-out prompts and negative controls will test specificity. Comparing isolated directions with interacting subspaces will test whether safety is localized or distributed across computation.

*Extending the analysis: Stability under adaptation.* I will track these mechanisms through controlled fine-tuning and adversarial perturbations, testing whether mechanism changes predict changes in vulnerability. The aim is to identify when a monitor or steering target requires recalibration and to develop stress tests that expose those limits. Comparisons across model families will identify which explanations generalize and which remain model-specific.

*Connecting the thrusts: Mechanistically informed intervention.* I will translate reproducible causal findings into monitoring signals and intervention targets, then test their contribution to Thrust I at matched resource budgets. This comparison will establish when mechanistic access adds value beyond behavioral monitoring. Held-out trajectories and independent outcome checks will distinguish genuine safety improvements from changes that only suppress monitor signals.

**Compute and laboratory strategy.** I will begin with existing open-weight models, at scales that permit repeated controlled experiments, and bounded tool-use environments. Cached trajectories, selective activation recording, and parameter-efficient adaptation will support repeated causal experiments on a small shared GPU pool. I will budget for activation storage, repeated seeds, and adversarial evaluation, since these can dominate costs even without pretraining. Larger-model replication will follow only after smaller-scale results justify it, through campus allocations or external compute support that I will seek. Model-scale transfer will be tested explicitly; findings from smaller models will not be presented as frontier-model guarantees.

Shared models and evaluation infrastructure will support student projects with distinct scientific questions. Building on my mentoring experience, I envision a research seminar in which students reproduce a safety claim, design a counterexample, and evaluate an intervention.

**Support and collaboration.** I would seek support through opportunities such as NSF CAREER and NSF SaTC 2.0, connecting safety research with student training and adversarial evaluation. Evaluation and standards partnerships, including potential engagement with NIST, would help connect controlled experiments to broader testing needs.

Through this program, I aim to make internal computation a practical basis for safety evaluation and control: explaining vulnerabilities, anticipating unsafe actions, and directing interventions whose benefits remain measurable as models and environments change.

## Selected References


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